Laboratory Work 5

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Course: Databases

Time: Wednesday 12:00-13:00

Report scope

This laboratory report covers CHECK and UNIQUE constraints, ticket discount validation, and default values in the airport_lab1 database.

CHECK constraints added with NOT VALID apply to new and updated rows without validating existing records. Defaults supply values when the corresponding columns are omitted from an INSERT.

Task 1 Minimum passenger age

Require passengers to be at least 10 years old and provide a date of birth.

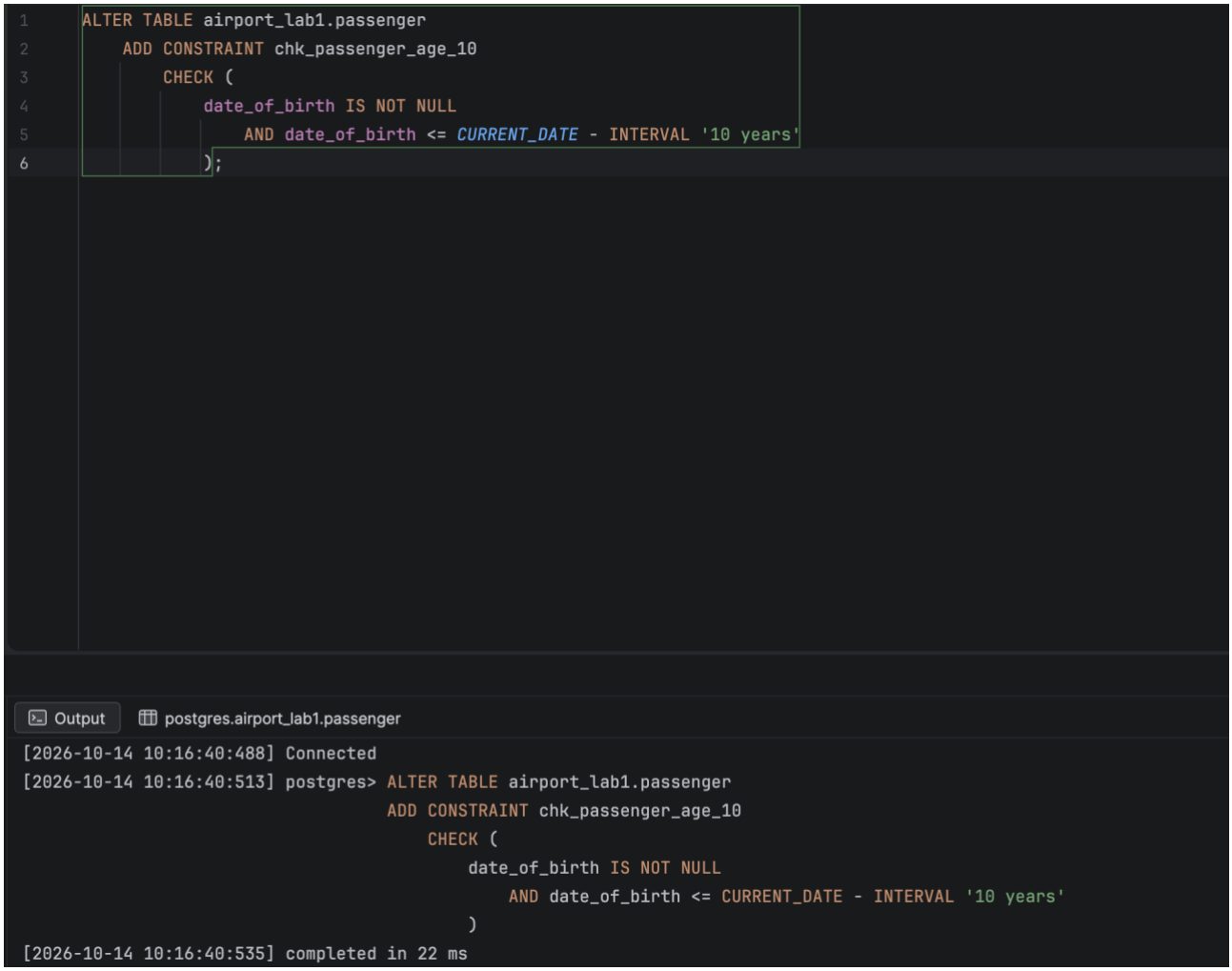


Figure 1. Passenger age check and execution output.

Task 2 Booking price range

Accept ticket prices greater than 0 and less than 50,000 KZT. Both boundaries are excluded.

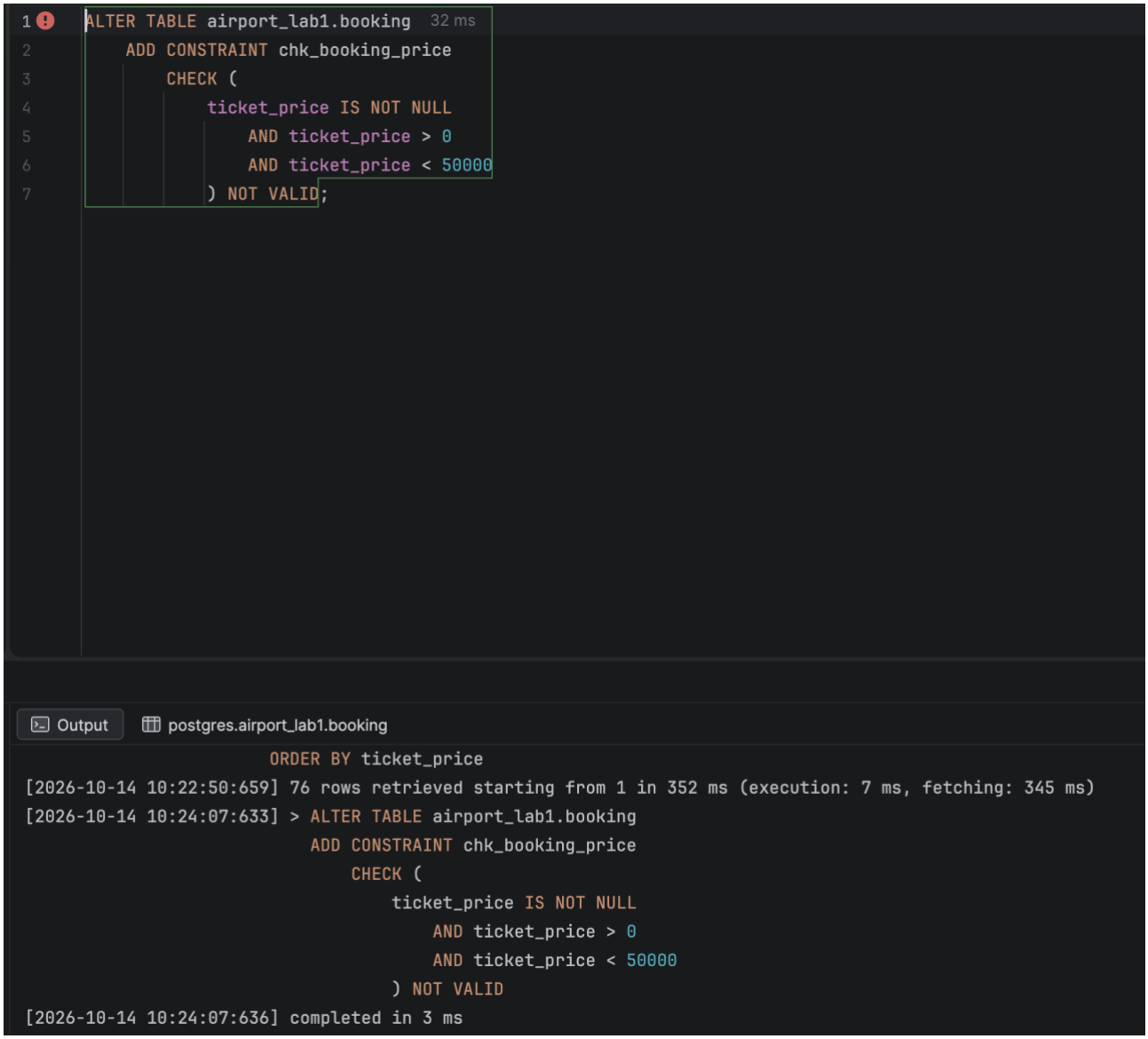


Figure 2. Booking price check added with NOT VALID.

Task 3 Baggage weight range

Accept baggage weights from 1 to 23 kg, including both boundaries.

```sql
ALTER TABLE airport_lab1.baggage
    ADD CONSTRAINT chk_baggage_weight
        CHECK (
            weight_kg IS NOT NULL
                AND weight_kg BETWEEN 1 AND 23
            );
```

Figure 3. SQL condition for the permitted baggage weight.

Task 4 Airport name length

Require airport names to contain at least 10 characters.

```sql
ALTER TABLE airport_lab1.airport  520 ms
    DROP CONSTRAINT IF EXISTS chk_airport_name_length,
    ADD CONSTRAINT chk_airport_name_length
        CHECK (
            airport_name IS NOT NULL
                AND CHAR_LENGTH(airport_name) >= 10
            ) NOT VALID;
```

Figure 4. Airport name length check.

Task 5 Unique constraints

Ensure that selected columns in each table have a UNIQUE constraint. Reuse equivalent existing constraints.

```
DO $$ 3 s
    DECLARE
        item RECORD;
        target_table REGCLASS;
        column_number SMALLINT;
    BEGIN
        FOR item IN
            SELECT *
            FROM (VALUES
                    ( table_name 'airport',          column_name 'airport_id'),
                    ( table_name 'airline',          column_name 'airline_code'),
                    ( table_name 'flight',           column_name 'flight_id'),
                    ( table_name 'passenger',        column_name 'passport_number'),
                    ( table_name 'booking',          column_name 'booking_id'),
                    ( table_name 'booking_change',   column_name 'booking_change_id'),
                    ( table_name 'boarding_pass',    column_name 'booking_id'),
                    ( table_name 'baggage',          column_name 'baggage_id'),
                    ( table_name 'baggage_check',    column_name 'baggage_check_id'),
                    ( table_name 'security_check',   column_name 'security_check_id')
                ) AS columns_to_check(table_name, column_name)
            LOOP
                target_table :=
                        FORMAT('airport_lab1.%I', item.table_name)::REGCLASS;

                SELECT attnum
                INTO STRICT column_number
                FROM pg_attribute
                WHERE attrelid = target_table
                  AND attname = item.column_name
                  AND NOT attisdropped;

                IF NOT EXISTS (
                    SELECT 1
                    FROM pg_constraint
                    WHERE conrelid = target_table
                      AND contype = 'u'
```

Figure 5a. Table and column selection and constraint lookup.

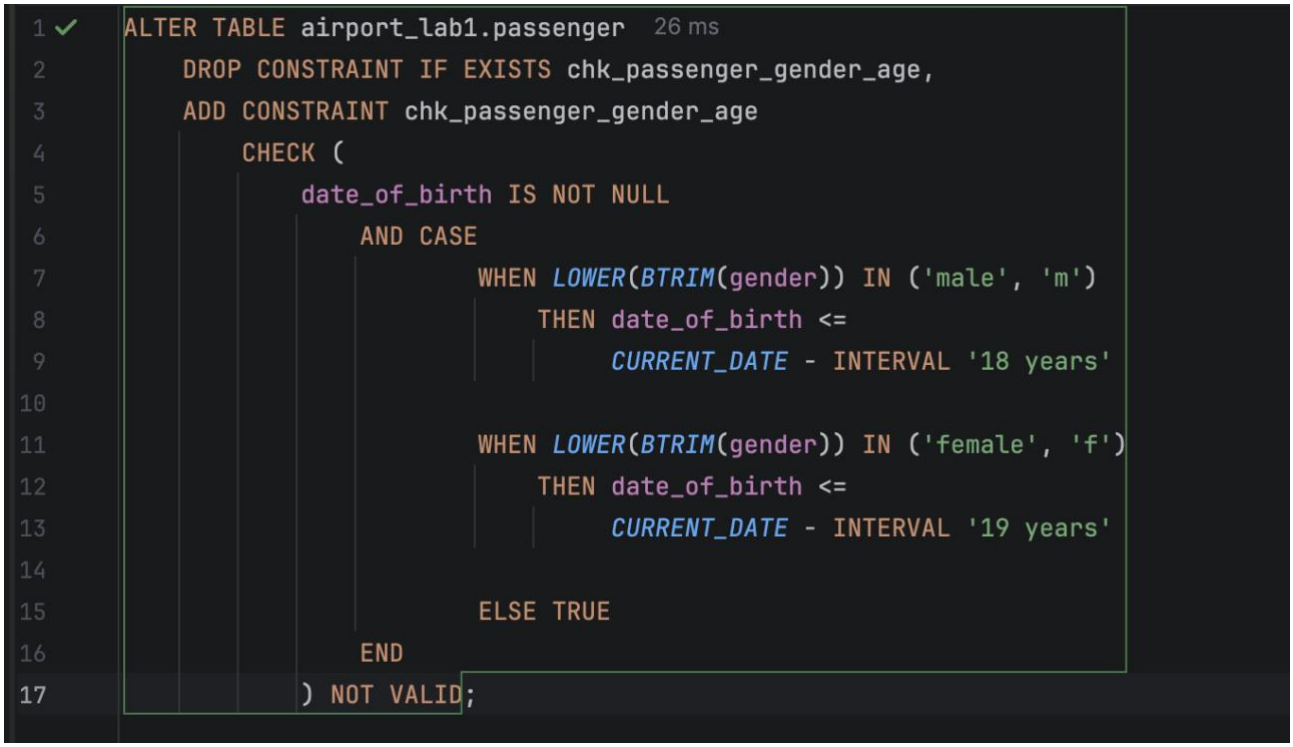
```
                      AND contype = 'u'
                      AND conkey = ARRAY[column_number]::SMALLINT[]
                ) THEN
                    EXECUTE FORMAT(
                            'ALTER TABLE %s ADD UNIQUE (%I)',
                            target_table,
                            item.column_name
                            );
                END IF;
            END LOOP;
    END;
$$;
```

Figure 5b. Conditional creation of missing UNIQUE constraints.

UNIQUE on an existing primary key is redundant; the script retains the selected identifiers used in this exercise.

Task 6 Minimum age by gender

Require male passengers to be at least 18 and female passengers to be at least 19 years old.



```
ALTER TABLE airport_lab1.passenger  26 ms
    DROP CONSTRAINT IF EXISTS chk_passenger_gender_age,
    ADD CONSTRAINT chk_passenger_gender_age
        CHECK (
            date_of_birth IS NOT NULL
                AND CASE
                        WHEN LOWER(BTRIM(gender)) IN ('male', 'm')
                            THEN date_of_birth <=
                                 CURRENT_DATE - INTERVAL '18 years'

                        WHEN LOWER(BTRIM(gender)) IN ('female', 'f')
                            THEN date_of_birth <=
                                 CURRENT_DATE - INTERVAL '19 years'

                        ELSE TRUE
                END
            ) NOT VALID;
```

Figure 6. Age conditions for male and female passengers.

Task 7 Minimum age by citizenship

Require a minimum age of 18 for Kazakhstan, 17 for France, and 19 for other countries.

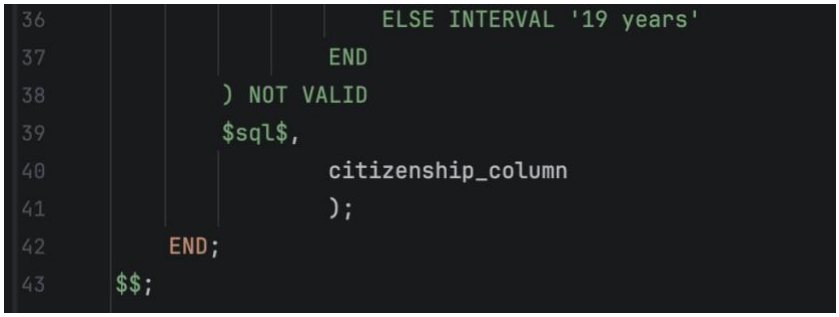


```
DO $$  35 ms
    DECLARE
        citizenship_column TEXT;
    BEGIN
        SELECT column_name
        INTO citizenship_column
        FROM information_schema.columns
        WHERE table_schema = 'airport_lab1'
          AND table_name = 'passenger'
          AND column_name IN (
                              'country_of_citizenship',
                              'citizenship_country'
            )
        ORDER BY CASE
                     WHEN column_name = 'country_of_citizenship' THEN 0
                     ELSE 1
                     END
        LIMIT 1;

        IF citizenship_column IS NULL THEN
            RAISE EXCEPTION 'В passenger не найден столбец гражданства';
        END IF;

        EXECUTE FORMAT(
                $sql$
        ALTER TABLE airport_lab1.passenger
        DROP CONSTRAINT IF EXISTS chk_passenger_citizenship_age,
        ADD CONSTRAINT chk_passenger_citizenship_age
        CHECK (
            date_of_birth IS NOT NULL
            AND %1$I IS NOT NULL
            AND date_of_birth <= CURRENT_DATE -
                CASE LOWER(BTRIM(%1$I))
                    WHEN 'kazakhstan' THEN INTERVAL '18 years'
                    WHEN 'france' THEN INTERVAL '17 years'
                    ELSE INTERVAL '19 years'
```

Figure 7a. Citizenship column lookup and country conditions.



```
                    ELSE INTERVAL '19 years'
                END
            ) NOT VALID
            $sql$,
                    citizenship_column
                    );
        END;
    $$;
```

Figure 7b. Completion of the citizenship constraint.

Task 8 Ticket discount validation

Store the discount amount in ticket_discount and check it against the price and creation date. Apply 5% on or after 2024-01-01 and 10% before that date.

```
ALTER TABLE airport_lab1.booking  20 ms
    ADD COLUMN IF NOT EXISTS ticket_discount NUMERIC(10, 2);

ALTER TABLE airport_lab1.booking  14 ms
    DROP CONSTRAINT IF EXISTS chk_booking_discount,
    ADD CONSTRAINT chk_booking_discount
        CHECK (
            created_at IS NOT NULL
                AND ticket_price IS NOT NULL
                AND ticket_discount IS NOT NULL
                AND ticket_discount = ROUND(
                    ticket_price *
                    CASE
                        WHEN created_at::date >= DATE '2024-01-01'
                            THEN 0.05
                        ELSE 0.10
                        END,
                    2
                                        )
            ) NOT VALID;
```

Figure 8. Discount column and CHECK constraint.

The CHECK validates a supplied discount amount. It does not calculate the value automatically. Existing rows are not backfilled.

Task 9 Default booking status

Set the default booking status to 'Confirmed' and allow this value in the status check.



```
DO $$  24 ms
    DECLARE
        old_constraint RECORD;
    BEGIN
        FOR old_constraint IN
            SELECT c.conname
            FROM pg_constraint c
                     JOIN pg_attribute a
                          ON a.attrelid = c.conrelid
                              AND a.attname = 'status'
            WHERE c.conrelid = 'airport_lab1.booking'::REGCLASS
              AND c.contype = 'c'
              AND c.conkey = ARRAY[a.attnum]::SMALLINT[]
            LOOP
                EXECUTE FORMAT(
                        'ALTER TABLE airport_lab1.booking DROP CONSTRAINT %I',
                        old_constraint.conname
                        );
            END LOOP;

        ALTER TABLE airport_lab1.booking
            ADD CONSTRAINT chk_booking_status
                CHECK (
                    LOWER(status) IN (
                                      nding',
                                     'confirmed',
                                     'cancelled',
                                     'checked_in'
                        )
                    ) NOT VALID;

        ALTER TABLE airport_lab1.booking
            ALTER COLUMN status SET DEFAULT 'Confirmed';
    END;
$$;
```

Figure 9. Status check replacement and default value configuration.

Task 10 Default creation date

Set created_at to the current date when the column is omitted from a new INSERT.



```
ALTER TABLE airport_lab1.booking  18 ms
    ALTER COLUMN created_at SET DEFAULT CURRENT_DATE;
```

Figure 10. Current date configured as the creation date default.